

LOSO (glove_ml_explorer) Experiment SB_ypr_accel

Static · YPR + accel (both hands), no Nothing

Generated 2026-05-12 03:40:33

Experiment configuration

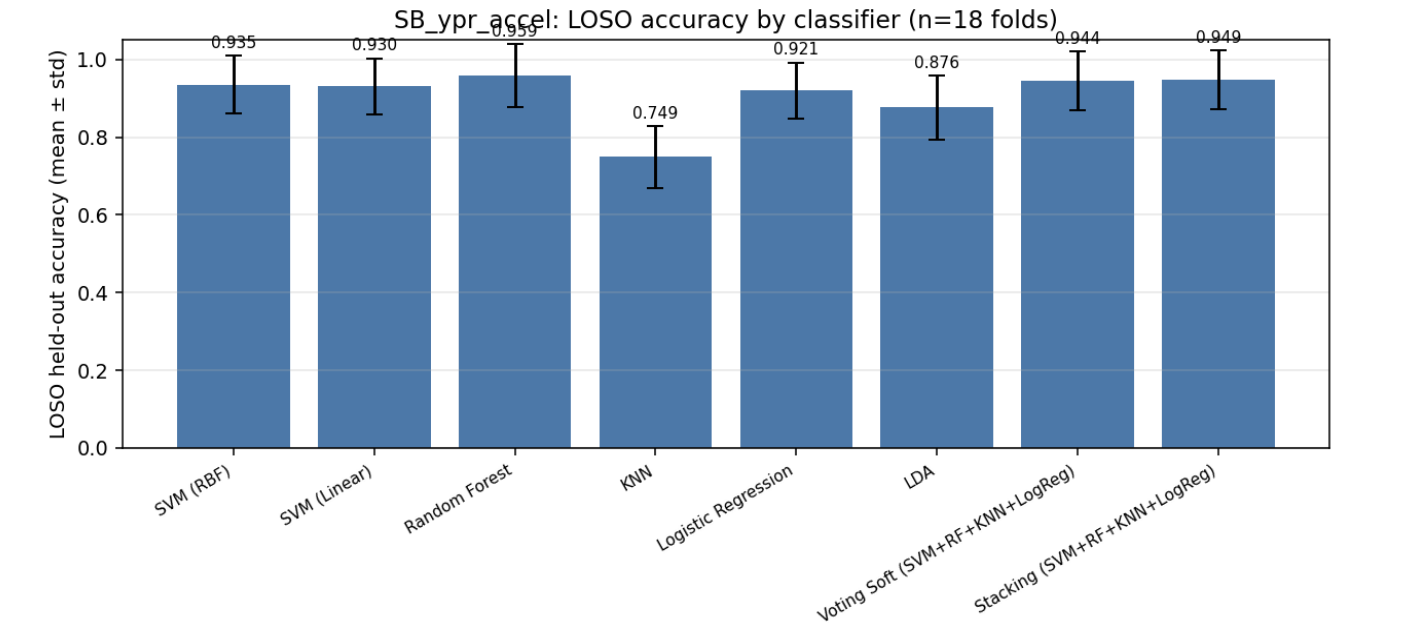
Class set	6 classes excluding Double_Nothing
Classes	Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman
Sensor modalities	YPR, accel
Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Sensor channels	132
Sequence length	90 steps (resampled)
Filter	off
Feature mode	stats+fft
Normalisation	minmax
Augmentation	off
Inner CV folds	5
LOSO subjects	18: 10_Josh, 11_Mansh, 12_Marcus, 13_Nat, 14_StephenV2, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 5_Georgia, 6_Harry, 7_Henry, 8_Jestin, 9_Joselyn, Stephen, Val

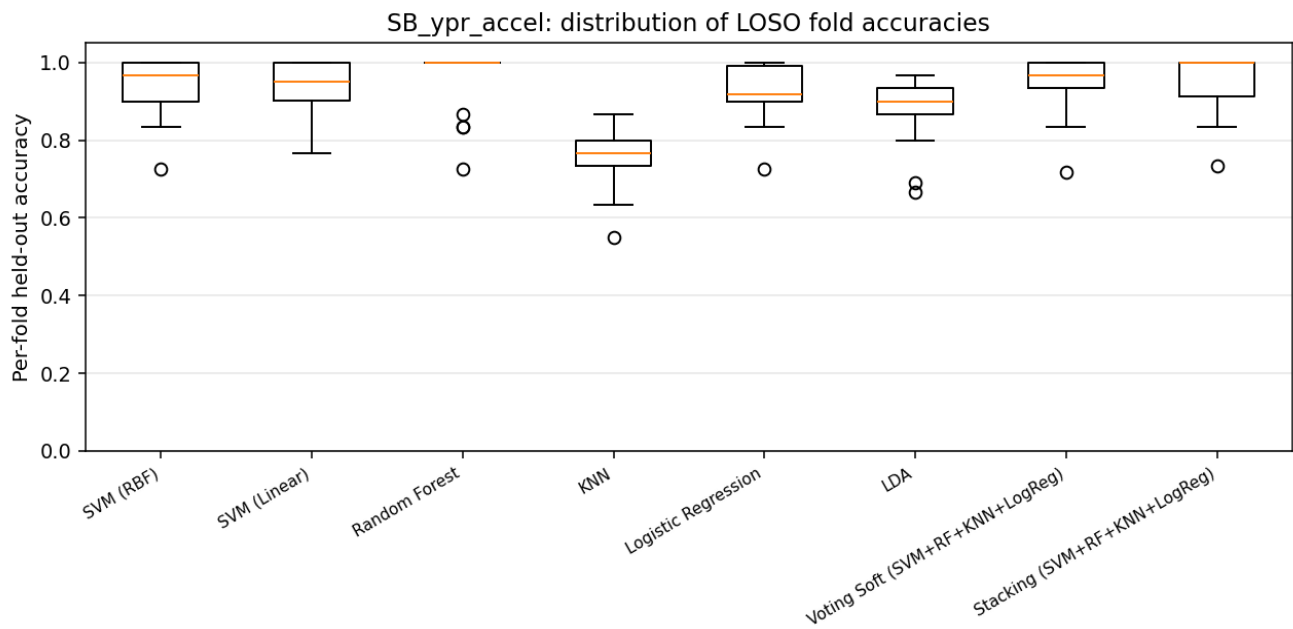
LOSO summary across all subjects

Classifier	Params*	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean
SVM (RBF)	—	0.9355	0.0737	0.7250	1.0000	0.9641
SVM (Linear)	106,935	0.9304	0.0728	0.7667	1.0000	0.9683
Random Forest	15,680	0.9588	0.0811	0.7250	1.0000	0.9667
KNN	—	0.7489	0.0789	0.5500	0.8667	0.7976
Logistic Regression	42,774	0.9207	0.0718	0.7250	1.0000	0.9652
LDA	42,774	0.8760	0.0821	0.6667	0.9667	0.9147
Voting Soft (SVM+RF+KNN+LogReg)	—	0.9442	0.0756	0.7167	1.0000	0.9743
Stacking (SVM+RF+KNN+LogReg)	—	0.9486	0.0753	0.7333	1.0000	0.9776

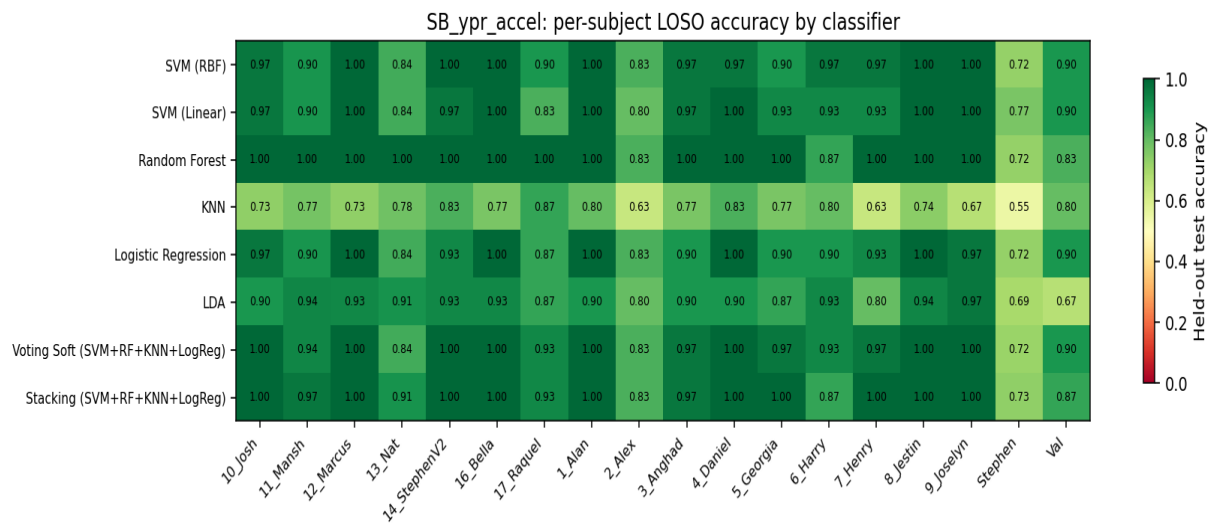
*Params: linear models use coef_+intercept size; tree ensembles use total node count across trees. SVM-RBF / KNN / Voting / Stacking are non-parametric or composite, so they show as —.

Best classifier by LOSO mean: **Random Forest**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Stacking (SVM+RF+KNN+LogReg)
10_Josh	604	0.967	0.967	1.000	0.733	0.967	0.900	1.000	1.000
11_Mansh	603	0.903	0.903	1.000	0.774	0.903	0.935	0.935	0.968
12_Marcus	604	1.000	1.000	1.000	0.733	1.000	0.933	1.000	1.000
13_Nat	602	0.844	0.844	1.000	0.781	0.844	0.906	0.844	0.906
14_StephenV2	604	1.000	0.967	1.000	0.833	0.933	0.933	1.000	1.000
16_Bella	604	1.000	1.000	1.000	0.767	1.000	0.933	1.000	1.000
17_Raquel	604	0.900	0.833	1.000	0.867	0.867	0.867	0.933	0.933
1_Alan	604	1.000	1.000	1.000	0.800	1.000	0.900	1.000	1.000
2_Alex	604	0.833	0.800	0.833	0.633	0.833	0.800	0.833	0.833
3_Anghad	604	0.967	0.967	1.000	0.767	0.900	0.900	0.967	0.967
4_Daniel	604	0.967	1.000	1.000	0.833	1.000	0.900	1.000	1.000
5_Georgia	604	0.900	0.933	1.000	0.767	0.900	0.867	0.967	1.000
6_Harry	604	0.967	0.933	0.867	0.800	0.900	0.933	0.933	0.867
7_Henry	604	0.967	0.933	1.000	0.633	0.933	0.800	0.967	1.000
8_Jestin	603	1.000	1.000	1.000	0.742	1.000	0.935	1.000	1.000
9_Joselyn	604	1.000	1.000	1.000	0.667	0.967	0.967	1.000	1.000
Stephen	514	0.725	0.767	0.725	0.550	0.725	0.692	0.717	0.733
Val	604	0.900	0.900	0.833	0.800	0.900	0.667	0.900	0.867

Classifier: SVM (RBF)

Params (best-effort): —

LOSO mean accuracy: 0.9355 ± 0.0737 (min 0.7250, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	5/5	5/5	5/5	5/5	5/5	0.967
11_Mansh	4/6	4/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	0/5	6/6	5/5	0.844
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	3/5	5/5	4/5	5/5	5/5	0.900
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	4/5	5/5	5/5	5/5	0.967
5_Georgia	5/5	3/5	4/5	5/5	5/5	5/5	0.900
6_Harry	5/5	5/5	5/5	4/5	5/5	5/5	0.967
7_Henry	5/5	4/5	5/5	5/5	5/5	5/5	0.967
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	5/20	18/20	13/20	15/20	18/20	18/20	0.725
Val	3/5	5/5	5/5	4/5	5/5	5/5	0.900
MEAN	0.895	0.928	0.958	0.842	0.994	0.994	0.935

Classifier: SVM (Linear)

Params (best-effort): 106,935

LOSO mean accuracy: 0.9304 ± 0.0728 (min 0.7667, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	5/5	5/5	5/5	5/5	5/5	0.967
11_Mansh	4/6	4/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	0/5	6/6	5/5	0.844
14_StephenV2	4/5	5/5	5/5	5/5	5/5	5/5	0.967
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	3/5	5/5	2/5	5/5	5/5	0.833
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	4/5	5/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	4/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	5/5	3/5	5/5	5/5	0.933
7_Henry	4/5	4/5	5/5	5/5	5/5	5/5	0.933
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	4/20	17/20	16/20	18/20	19/20	18/20	0.767
Val	3/5	5/5	4/5	5/5	5/5	5/5	0.900
MEAN	0.859	0.936	0.967	0.828	0.997	0.994	0.930

Classifier: Random Forest

Params (best-effort): 15,680

LOSO mean accuracy: 0.9588 ± 0.0811 (min 0.7250, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	6/6	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	5/5	5/5	5/5	5/5	5/5	5/5	1.000
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	1/5	5/5	5/5	0.867
7_Henry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	3/20	17/20	20/20	12/20	17/20	18/20	0.725
Val	1/5	5/5	5/5	4/5	5/5	5/5	0.833
MEAN	0.908	0.992	1.000	0.867	0.992	0.994	0.959

Classifier: KNN

Params (best-effort): —

LOSO mean accuracy: 0.7489 ± 0.0789 (min 0.5500, max 0.8667)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	3/5	4/5	1/5	5/5	5/5	0.733
11_Mansh	3/6	3/5	5/5	3/5	5/5	5/5	0.774
12_Marcus	4/5	3/5	5/5	0/5	5/5	5/5	0.733
13_Nat	6/6	5/5	3/5	0/5	6/6	5/5	0.781
14_StephenV2	5/5	5/5	3/5	2/5	5/5	5/5	0.833
16_Bella	3/5	4/5	4/5	2/5	5/5	5/5	0.767
17_Raquel	5/5	3/5	5/5	3/5	5/5	5/5	0.867
1_Alan	5/5	3/5	4/5	2/5	5/5	5/5	0.800
2_Alex	5/5	2/5	2/5	0/5	5/5	5/5	0.633
3_Anghad	4/5	4/5	2/5	3/5	5/5	5/5	0.767
4_Daniel	5/5	5/5	4/5	1/5	5/5	5/5	0.833
5_Georgia	3/5	4/5	5/5	1/5	5/5	5/5	0.767
6_Harry	5/5	4/5	2/5	3/5	5/5	5/5	0.800
7_Henry	4/5	3/5	0/5	2/5	5/5	5/5	0.633
8_Jestin	5/6	5/5	3/5	0/5	5/5	5/5	0.742
9_Joselyn	2/5	4/5	4/5	0/5	5/5	5/5	0.667
Stephen	3/20	15/20	8/20	5/20	17/20	18/20	0.550
Val	2/5	2/5	5/5	5/5	5/5	5/5	0.800
MEAN	0.760	0.731	0.689	0.325	0.992	0.994	0.749

Classifier: Logistic Regression

Params (best-effort): 42,774

LOSO mean accuracy: 0.9207 ± 0.0718 (min 0.7250, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	5/5	5/5	5/5	5/5	5/5	0.967
11_Mansh	4/6	4/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	0/5	6/6	5/5	0.844
14_StephenV2	4/5	5/5	4/5	5/5	5/5	5/5	0.933
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	3/5	5/5	3/5	5/5	5/5	0.867
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	3/5	5/5	4/5	5/5	5/5	5/5	0.900
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	4/5	4/5	5/5	5/5	0.900
6_Harry	5/5	5/5	5/5	2/5	5/5	5/5	0.900
7_Henry	4/5	4/5	5/5	5/5	5/5	5/5	0.933
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	4/5	5/5	5/5	5/5	5/5	5/5	0.967
Stephen	3/20	17/20	16/20	15/20	18/20	18/20	0.725
Val	3/5	5/5	5/5	4/5	5/5	5/5	0.900
MEAN	0.845	0.936	0.956	0.797	0.994	0.994	0.921

Classifier: LDA

Params (best-effort): 42,774

LOSO mean accuracy: 0.8760 ± 0.0821 (min 0.6667, max 0.9667)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	3/5	5/5	4/5	5/5	5/5	5/5	0.900
11_Mansh	5/6	5/5	5/5	5/5	4/5	5/5	0.935
12_Marcus	5/5	4/5	5/5	4/5	5/5	5/5	0.933
13_Nat	6/6	4/5	5/5	3/5	6/6	5/5	0.906
14_StephenV2	3/5	5/5	5/5	5/5	5/5	5/5	0.933
16_Bella	4/5	5/5	5/5	4/5	5/5	5/5	0.933
17_Raquel	5/5	4/5	5/5	2/5	5/5	5/5	0.867
1_Alan	4/5	4/5	5/5	4/5	5/5	5/5	0.900
2_Alex	4/5	5/5	4/5	1/5	5/5	5/5	0.800
3_Anghad	3/5	5/5	5/5	4/5	5/5	5/5	0.900
4_Daniel	4/5	5/5	5/5	3/5	5/5	5/5	0.900
5_Georgia	4/5	5/5	5/5	2/5	5/5	5/5	0.867
6_Harry	5/5	5/5	5/5	3/5	5/5	5/5	0.933
7_Henry	4/5	5/5	3/5	3/5	5/5	4/5	0.800
8_Jestin	5/6	4/5	5/5	5/5	5/5	5/5	0.935
9_Joselyn	4/5	5/5	5/5	5/5	5/5	5/5	0.967
Stephen	1/20	15/20	16/20	15/20	18/20	18/20	0.692
Val	2/5	4/5	1/5	4/5	4/5	5/5	0.667
MEAN	0.751	0.919	0.900	0.731	0.972	0.983	0.876

Classifier: Voting Soft (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9442 ± 0.0756 (min 0.7167, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	4/6	5/5	5/5	5/5	5/5	5/5	0.935
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	0/5	6/6	5/5	0.844
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	4/5	5/5	4/5	5/5	5/5	0.933
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	5/5	5/5	5/5	5/5	0.967
6_Harry	5/5	5/5	5/5	3/5	5/5	5/5	0.933
7_Henry	5/5	4/5	5/5	5/5	5/5	5/5	0.967
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	3/20	18/20	15/20	14/20	18/20	18/20	0.717
Val	3/5	5/5	5/5	4/5	5/5	5/5	0.900
MEAN	0.901	0.961	0.986	0.828	0.994	0.994	0.944

Classifier: Stacking (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9486 ± 0.0753 (min 0.7333, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
11_Mansh	5/6	5/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
13_Nat	6/6	5/5	5/5	2/5	6/6	5/5	0.906
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	4/5	5/5	4/5	5/5	5/5	0.933
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	2/5	4/5	5/5	0.867
7_Henry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	3/20	17/20	15/20	17/20	18/20	18/20	0.733
Val	2/5	5/5	5/5	4/5	5/5	5/5	0.867
MEAN	0.899	0.981	0.986	0.847	0.983	0.994	0.949